



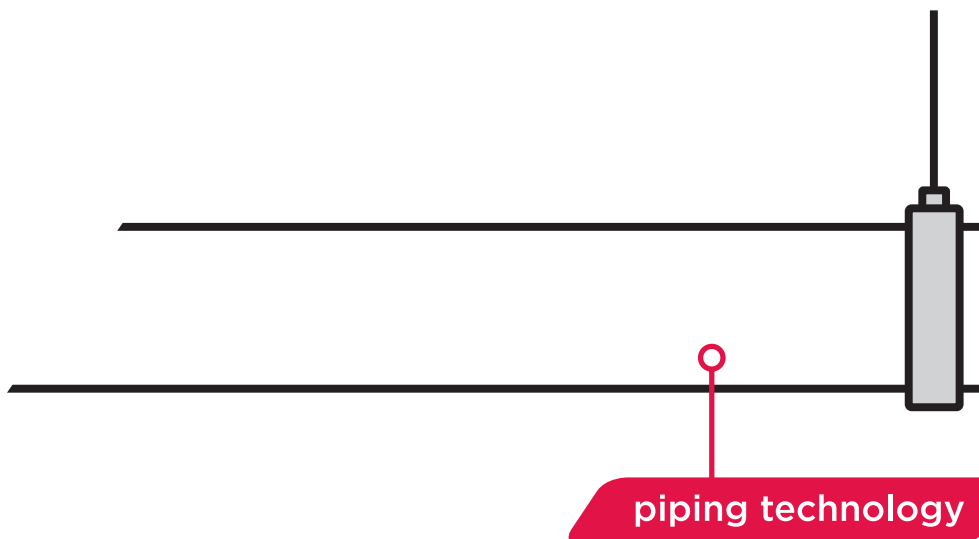


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Aalberts integrated piping systems

don't just buy
products,
buy solutions.





SHURJOINT®

a world leader in mechanical piping components

Mechanical couplings were first developed in the 1920's and evolved into Grooved Mechanical Couplings during the 1950's & 60's. This revolution was further spurred in the latter half of the 20th century by advanced engineering and innovative materials such as Ductile Iron and EPDM elastomers.

past to present

The history of Shurjoint dates back to 1974, when the founders produced their first grooved couplings. These first couplings were produced from malleable iron, the casting material of choice at this time. Before long foundry production was converted to ductile iron. Ductile iron was the ideal material, providing for precision castings and superior strength, without the need for further heat treatment. EPDM (ethylene propylene diene monomer) provided a great advancement in synthetic rubber elastomer compounds. This new compound offered a service life equal to or in some cases longer than that of carbon steel pipe. EPDM was an ideal gasket material for grooved piping systems.

a world leader

With over four decades of experience, Shurjoint is recognized as a world leader in the design and manufacture of mechanical piping components. Shurjoint has developed and currently offers over 3000 individual piping components in sizes from ½" to 104", for use with a variety of piping materials including carbon steel, stainless steel, ductile iron, PVC, HDPE, CPVC and copper tubing.

our mission

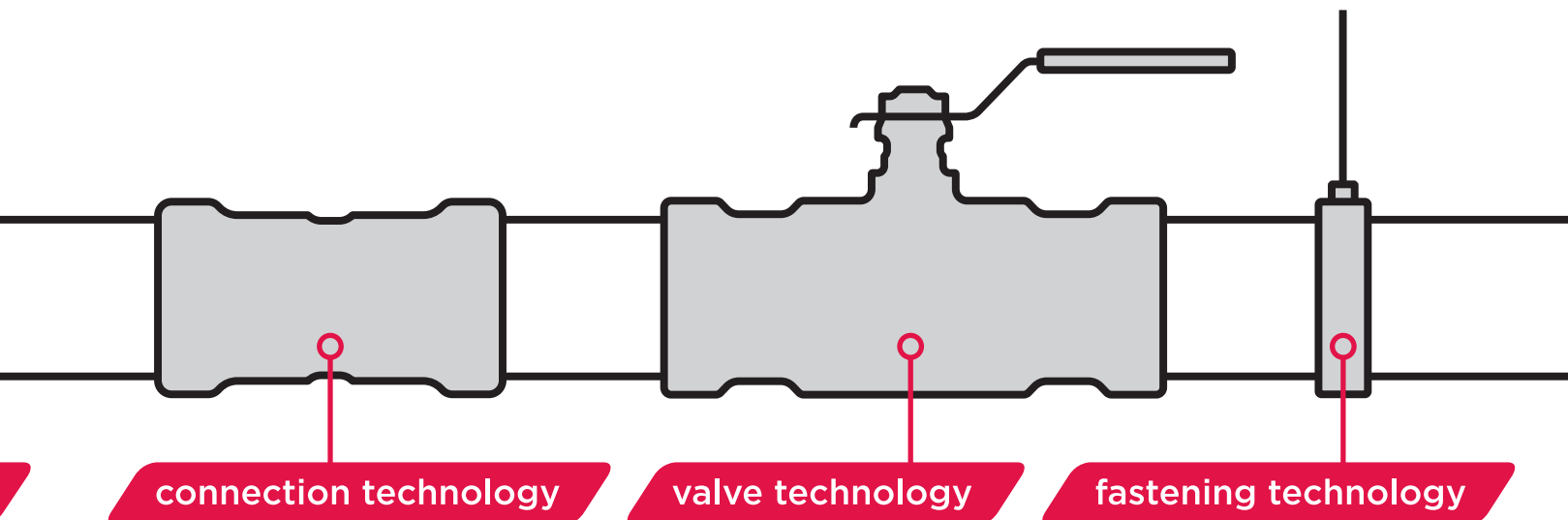
Our mission is to supply the highest quality products to customers worldwide with an unmatched level of customer service at a superior value. In addition to these hallmarks, Shurjoint continuously invests in research, engineering and development, resulting in innovative products and new solutions for the changing needs of industry.

this catalog

This catalog features our general product offering. For the latest and most complete listing of products, news and additional information please visit our website www.shurjoint.com or contact our Customer Service Representative.

applications

- HVAC
- fire protection
- water supply & treatment
- fire plumbing
- municipal
- food processing
- pulp & paper
- agriculture
- reverse osmosis
- desalination
- mining & tunnel boring
- marine
- gas
- chemical
- oil
- air





general information

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general notes

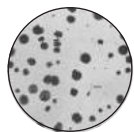
1. Always read and understand all Shurjoint installation instructions before installing any Shurjoint product.
2. Always depressurize and drain the piping system before attempting disassembly, adjustment or removal of any piping components.
3. Designers must know and understand all relevant building and or piping standards, codes and specifications. It is the responsibility of the designer to select and or specify the appropriate product for the intended use and service.
4. Always refer to the maximum pressure rating and service temperature range allowed for Shurjoint products and ensure that they are used within these parameters.
5. Special attention is required for selection of suitable gasket grades for the intended service application.
6. All information and data contained herein supersede all previous published data. Shurjoint reserves the right to change product designs and or specifications without notice and or obligation. Please refer to the Shurjoint web site for the latest information.

ductile iron - housing material

Ductile iron is an ideal material for grooved mechanical components, as it provides similar or greater strength to that of wrought or cast steel piping materials such as; forged steel flanges - ASTM A105, carbon steel valves - ASTM A216 WCB, wrought carbon steel pipe - ASTM A53 Gr. B, etc. Most Shurjoint components are made of ductile iron conforming to ASTM A536 Gr. 65-45-12 and or ASTM A395 Gr. 65-45-15.

ductile iron

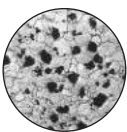
Ductile iron was first invented in the U.S.A. and U.K. in the late 1940's. Superior strength was achieved by crystallizing graphite in the shape of nodules. The result was ductile iron that had tensile and yield strength properties that were equal to or greater than some steel castings. This superior strength combined with ductile iron's excellent castability helped to reduce the weight and cost of many components. Because of these advantages and benefits, many components have been converted from gray iron, malleable iron and steel castings to ductile iron over the past 80 years. Please visit the Ductile Iron Society website; www.ductile.org, for further information.



ductile iron
superior tensile strength
with good castability



gray iron
excellent castability but
'brittle' - less strength



malleable iron
stronger than gray iron
but poor castability

International ductile iron specifications equivalent to ASTM A536 Gr. 65-45-12 and or ASTM A395 Gr. 65-45-15 are:

- SAE J434: D4512
- EN1563: EN-GJS-450-10 or EN-GJS-450-15
- JIS G5502: FCD450-10
- SABS 936/937: SG42

physical strength of materials comparative

	ASTM designation	tensile strength, min. psi (MPa)	yield strength, min. psi (MPa)	elongation in 2", %
ductile iron castings	A536: Gr. 65-45-12	65,000 (448)	45,000 (310)	12
ductile iron castings	A395: Gr. 65-45-15	65,000 (448)	45,000 (310)	15
forged carbon steel	A105	70,000 (485)	40,000 (250)	20
cast carbon steel	A216: WCB	70,000 (485)	36,000 (205)	22
carbon steel pipe	A53: Gr. B	60,000 (415)	35,000 (240)	(29.5)
malleable iron castings	A47: Gr. 32510	51,000 (345)	32,000 (224)	10
gray iron castings	A126: Gr. B	31,000 (214)	not specified	not specified

ASTM A536, grade 65-45-12 (UNS F33100)

chemical requirements*	min.	max.
carbon, %	3.0	3.9
silicon, %	2.5	3.0
manganese, %	0.1	0.4
phosphorus, %		0.07
sulfur, %		0.02
magnesium, %	0.03	0.05
chromium, %		0.1

physical properties	min.	max.
tensile strength, psi (MPa)	65,000 (448)	-
yield strength, psi (MPa)	45,000 (310)	-
elongation, %	12	-

*Reference only as chemical requirements are not specified in ASTM A536.

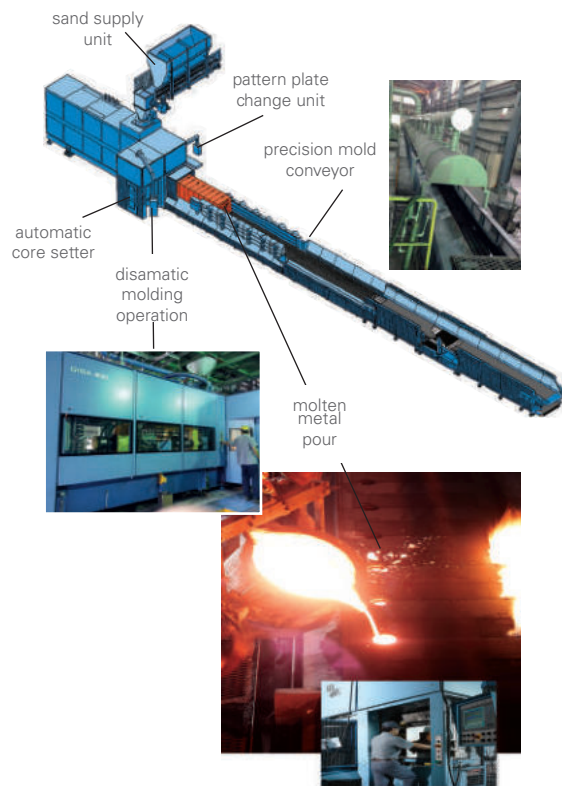
ASTM A395, grade 65-45-15 (UNS F33100)

chemical requirements	min.	max.
carbon, %	3.0	
silicon, %		2.5
manganese, %	not specified	
phosphorus, %		0.08
sulfur, %	not specified	
magnesium, %	not specified	
chromium, %	not specified	

physical properties	min.	max.
tensile strength, psi (MPa)	65,000 (448)	-
yield strength, psi (MPa)	45,000 (310)	-
elongation, %	12	-

disamatic molding line

max. 360 molds per hour



bolts & nuts

carbon steel bolts & nuts

Shurjoint products utilize oval neck track bolts conforming to ASTM A449 or ASTM A183 Gr. 2 and heavy duty nuts to ASTM A563 Gr. B, available with UNC threads or ISO metric threads. The UNC track bolts and nuts are supplied electro zinc plated in a silver chromate color and ISO metric bolts and nuts in a gold chromate color. Hot-dip galvanized bolts and nuts are also available upon request.

ASTM A449, quenched and tempered steel bolts*

chemical requirements*	min.	max.
carbon, %	0.28	0.55
manganese, %	0.60	
phosphorus, %		0.040
sulfur, %		0.050

physical properties	min.	max.
tensile strength, psi (MPa)	120,000 (825)	-
yield strength, psi (MPa)	92,000 (635)	-
elongation, %	14	-

*Equivalent to property class 8.8 bolts per ISO 898.

ASTM A183, grade 2 carbon steel track bolts

chemical requirements	min.	max.
carbon, %	0.30	
phosphorus, %		0.05
sulfur, %		0.06

physical properties	min.	max.
tensile strength, psi (MPa)	110,000 (760)	-
yield strength, psi (MPa)	80,000 (550)	-
elongation, %	12	-

ASTM A183, grade 2 carbon steel track bolts

chemical requirements (bolts)	min.	max.
carbon, %		0.55
phosphorus, %		0.12
sulfur, %		0.15

physical properties	min.	max.
hardness, rockwell	B69	C32

stainless steel bolts & nuts

Stainless steel track bolts and nuts, type 304 or type 316, are supplied with Shurjoint stainless steel couplings. Track bolts and nuts are molybdenum disulfide (MoS₂) coated to inhibit galling. As an option, silicon bronze nuts are also available to further reduce the chance of galling.

ASTM A193, grade B8 (type 304) stainless steel bolts

chemical requirements*	min.	max.
carbon, %		0.08
manganese, %		2.00
phosphorus, %		0.045
sulfur, %		0.030
silicon, %		1.00
chromium, %	18.00	20.00
nickel, %	8.00	10.50

physical properties	min.	max.
tensile strength, psi (MPa)	75,000 (515)	-
yield strength, psi (MPa)	30,000 (205)	-
elongation, %	30	-



a stainless steel bolt fastened with a silicon bronze nut

ASTM A193, grade B8M (type 316) stainless steel bolts

chemical requirements*	min.	max.
carbon, %		0.08
manganese, %		2.00
phosphorus, %		0.045
sulfur, %		0.030
silicon, %		1.00
chromium, %	16.00	18.00
nickel, %	10.00	14.00
molybdenum	2.00	3.00

physical properties	min.	max.
tensile strength, psi (MPa)	75,000 (515)	-
yield strength, psi (MPa)	30,000 (205)	-
elongation, %	30	-

ASTM B98 alloy B copper-silicon alloy (UNS no. C65100)

chemical requirements*	min.	max.
copper, %	96.0	
lead, %		0.05
iron, %		0.80
zinc, %		1.50
magnesium, %		0.70
silicon, %	0.8	2.00

physical properties	min.	max.
tensile strength, psi (MPa)	55,000 (380)	-
yield strength, psi (MPa)	20,000 (140)	-
elongation, %	11	12

recommended bolt torque range

Always use factory supplied bolts and nuts for assembly of Shurjoint couplings. Shown below are the general recommended torque ranges for common sizes of carbon steel bolts. Never exceed the recommended torque range by more than 25% as excessive torque can lead to joint failure, personal injury and/or property damage. Always depressurize and drain the piping system before attempting disassembly, adjustment or removal of any piping component. Follow installation instructions for proper assembly of all Shurjoint components. For questions contact Shurjoint.

bolt size		proper torque range	
mm	in	nm	lbs-ft
M10	3/8"	40-55	30-40
M12	1/2"	120-140	90-105
M16	5/8"	135-175	100-130
M20	3/4"	200-270	150-200
M22	7/8"	240-300	180-220
M24	1"	270-305	200-225
-	1 1/8"	-	225-275

For stainless steel bolts, reduce by 20%

rubber gasket compounds

The 20th century was the era of innovation in plastic and rubber materials. Among the new synthetic rubber compounds that most impacted our industry were EPDM (ethylene propylene diene monomer) and Nitrile rubbers. Note: please refer to the Gasket Selection Guide beginning on page 196 for additional information relating to service temperatures and chemical resistance.

EPDM

EPDM is recognized as the most water resistant rubber available today. Good for cold & hot water up to 230°F (110°C), waste water, water with acid, deionized water and seawater. EPDM is not recommended for use with petroleum based oils and fuels, hydrocarbon solvents and aromatic hydrocarbons.

Shurjoint Grade "E" EPDM is compounded per ASTM D2000 designation 2CA615A25B24F17Z. Peroxide curing and post curing give a higher crosslink density, which provides a higher aging resistance than required in AWWA C606.



basic requirements	AWWA C606 2CA615A25B24F17Z	Shurjoint standard
hardness, durometer A, point	65±7	60±5
tensile strength, psi, min.	1500 psi (10.34 MPa)	1500 psi (10.34 MPa)
elongation, %, min.	300 %	300 %
heat aging properties	after aged at 212°F (100°C) for 70 hours	after aged at 257°F (125°C) for 70 hours
change in durometer hardness, max.	+10 point	+5 points
change in tensile strength, max.	-25%	-10%
change in ultimate elongation, max.	-25%	-20%
compression set, method b, max.	25%	20%

Use Shurjoint Grade "E-pw" for potable water and food processing services. The Grade "E-pw" is UL classified per NSF/ANSI 61 and NSF/ANSI 372 for cold +86°F (30°C) and hot +180°F (82°C) potable water services. EPDM seals are recommended for use in breweries as they have the least impact on the characteristics of beer or wort.

note: EPDM materials used in domestic water applications with high levels of chlorine and or chloramines should be subjected to resistance testing, as not all materials will be suitable. EPDM materials with higher saturated ethylene content and lower carbon black content are recommended for chloramine and chlorine resistance. Contact Shurjoint for further information.

NBR, buna-n, and nitrile

All represent the same copolymer of butadiene and acrylonitrile (ACN), which is inherently resistant to hydraulic fluids, lubricating oils, transmission fluids and other non-polar petroleum based products and water less than 150° F (65° C). The higher the ACN content, the higher the resistance to oils and heat, but the lower elastic characteristics and compression set. NBR displays poor resistance to hot water and steam.



Shurjoint grade "T" NBR rubber is compounded based on ASTM D2000 designation 5BG615A14B24Z and exceeds the requirements of AWWA C606. Grade "T" is a general purpose compound with a medium ACN level. For fuels, especially those with a low aniline point, such as premium or unleaded gasoline, ASTM referenced fuels B & C and naphtha, use Shurjoint grade "M2" Epichloro-Hydrin or grade "O" Fluorocarbon.

basic requirements	AWWA C606 G615A14B24Z	Shurjoint standard
hardness, durometer A, point	60±7	60±5
tensile strength, psi, min.	1500 psi (10.34 MPa)	1500 psi (10.34 MPa)
elongation, %, min.	300 %	300 %
when heat aged at 212°F (100°C) for 70 hours	-	-
change in durometer hardness, max.	+10 point	+5 points
change in tensile strength, max.	-25%	-20%
change in ultimate elongation, max.	-30%	-30%
compression set, method b, max.	25%	25%

Use Shurjoint Grade "A" white Nitrile gaskets for oily and greasy food products and processing, as well as pharmaceutical and cosmetics manufacturing. The Grade "A" is compounded from FDA approved ingredients (CFR Title 21 Part 177.2600).



Use Shurjoint Grade "S" Nitrile gaskets for joints with AWWA ductile iron pipe. Good for mineral oils, vegetable oils, air with oil vapors and water less than 150°F (65°C).

silicone (VMQ)

Shurjoint Grade "L" Silicone compound features high temperature range stability and low temperature flexibility. Recommended for dry heat and air without hydrocarbons up to 350°F (177°C). Silicone compounds are used in many food and medical applications as they do not impart odor or taste. Not recommended for hot water or steam services.



red gasket

chloroprene (CR, neoprene)

Shurjoint Grade "V" chloroprene rubber is a general purpose elastomer that demonstrates good resistance to lubricating oils, animal & vegetable fats and greases. Chloroprene is not effective in aromatic and oxygenated solvent environments and is not recommended for hot water and steam services



yellow stripe

fluorocarbon (FKM)

FKM is a highly fluorinated carbon backbone compound and offers excellent resistance to harsh chemical and ozone attack with a thermal stability to 300°F (149°C). Shurjoint Grade "O" fluorocarbon gasket is recommended for use with oils, gasoline, hydraulic fluids, hydrocarbon solvents and extended fuels that fall outside the service parameters of grade T / NBR compounds. Not recommended for steam services.



blue stripe

epichloro-hydrin (ECO)

Shurjoint Grade "M2" compound offers good to excellent resistance to aliphatic hydrocarbon and aromatic hydrocarbon fuels at low temperatures, LP gases & fuels, mineral oils and many solvents. ECO offers limited resistance to many organic chemicals.



white stripe

halogenated butyl (CIIR)

Shurjoint Grade "M" CIIR is specially compounded for use with AWWA ductile iron pipe for water services, mild dilute acids, oil-free air and many chemicals. The compound is UL classified for potable water use per NSF/ANSI 61 and NSF/ANSI 372.



brown stripe

data chart notes

nom. size	pipe O.D.	max working pressure (CWP)	max. end load (CWP)	axial displacement	angular movement	dimensions				bolt size	bolt torque	weight
					degree per coupling	pipe	A	B	C			
in	in	psi	lbs	in	(°)	ft / in	in	in	in	in	lbs-ft	lbs
DN	mm	bar	kN	in / mm		mm / m	mm	mm	mm		nm	kgs
1	2	3	4	5	6	7	8	9	10			

1. nominal size: Shurjoint couplings and fittings are identified by the nominal IPS pipe size in inches or nominal diameter of pipe (DN) in millimeters. Refer to the chart on the next page which shows a comparison between typical metric and IPS pipe sizes

2. pipe O.D.: Actual outside diameter of pipe in inches and millimeters.

3. maximum working pressure (CWP): Maximum working pressures listed are CWP (cold water pressure) or maximum allowed working pressure within the service temperature range of the gasket used in the coupling, based on standard wall or sch. 40 steel pipe, cut or roll-grooved to ANSI/AWWA C606 (latest edition) specifications and tested to ASTM F1476. Burst test pressures are minimum 3 times the maximum working pressures unless otherwise specified. These ratings may occasionally differ from maximum working pressures listed and/or approved by UL, ULC, and/or FM as testing conditions and test pipes differ. For performance data on other pipe schedules contact Shurjoint.

note: For one time field test only the maximum joint working pressure may be increased 1½ times the figures shown.

4. maximum end load: Maximum end loads listed are total of internal and external forces to which the joint can be subjected, based on standard wall or sch. 40 steel pipe, cut or roll-grooved to ANSI/AWWA C606 (latest edition) specifications.

5. axial displacement: Designed range of the gap between pipe ends based on roll grooved pipe.

6. angular movement (deflection): Allowable Axial Displacement and Angular Movement (deflection) figures are for roll grooved standard steel pipe. Values for cut grooved pipe will be double that of roll grooved. These values are maximums; for design and installation purposes these figures should be reduced by: 50% for ¾"/DN20 – 3½"/DN90; 25% for 4"/DN100 and larger to compensate for jobsite conditions.

Maximum allowable deflection of pipe from centerline when the joint is used with cut or roll-grooved steel pipe under no internal pressure.

7. dimensions: "A", "B", "C" and so on are external dimensions for reference purpose only in inches and millimeters.

8. bolt size: UNC bolt size and length in inches and ISO metric bolt size and length in millimeters with numbers of bolts where applicable.

9. bolt torque: Recommended bolt fastening torque in Nm and Lbs-Ft.

10. weight: Weight of a coupling complete with gasket, bolts and nuts or of a fitting in kilograms and pounds.

nominal size / pipe o.d.

nominal size

While Shurjoint fittings are normally identified by the nominal size, always check the actual O.D. of the pipe and fittings to be connected, as in some markets it is customary to refer to different O.D. pipes with the same nominal size. For example: The nominal size 65 (2½") is referred to 73.0 mm (2.875") pipe O.D. in IPS and 76.1 mm (3.000") pipe O.D. in BS, ISO or JIS pipes. Refer to pipe & tubing standards for details.

pipe & tubing standards

(ANSI) American National Standards Institute B36.10 & B36.19
(API) American Petroleum Institute API 5L
(ASTM) American Society of Testing and Materials A135, A795 & B88
(BS) British Standards BS1387, BS3600, BS3601 & BS3605
(ISO) International Standard Organization 65 & 4200
(JIS) Japan Industrial Standard G3452 & G3459

global pipe size designations

Shurjoint product data & technical data are identified by the nominal IPS pipe size in inches or nominal diameter of pipe (DN) in millimeters.

The following chart shows a comparison between typical IPS size and metric (DIN) sizes.

nominal size			outside diameter (O.D.)							
inches (imperial)	DN (metric, mm)	(actual pipe O.D.)	DIN	BS	ISO	JIS	ANSI	GB	india	
									IS 1239	IS 3589
½	15	21.3 mm	DN 15	DN 15	DN 15	21.7 mm	½"	DN 15	DN 15	-
¾	20	26.7 mm	DN 20	DN 20	DN 20	27.2 mm	¾"	DN 20	DN 20	-
1	25	33.4 mm	DN 25	DN 25	DN 25	34.0 mm	1"	DN 25	DN 25	-
1¼	32	42.2 mm	DN 32	DN 32	DN 32	42.7 mm	1¼"	DN 32	DN 32	-
1½	40	48.3 mm	DN 40	DN 40	DN 40	48.6 mm	1½"	DN 40	DN 40	-
2	50	60.3 mm	DN 50	DN 50	DN 50	60.5 mm	2"	DN 50	DN 50	-
2 ½	65	73.0 mm	-	-	-	-	2½"	-	-	-
		76.1 mm BS/ISO	76.1 mm	76.1 mm	76.1 mm	76.3 mm	-	76.1 mm**	76.1 mm	-
3	80	88.9 mm	DN 80	DN 80	DN 80	DN 80	3"	DN 80	DN 80	-
3½	90	101.6 mm	-	-	-	-	-	-	-	-
4	100	108.0 mm China (& old DIN)	DIN 133.0 mm	-	-	-	-	108.0 mm**	-	-
		114.3 mm	DN 100	DN 100	DN 101	DN 100	4"	DN 100	DN 100	-
-	127.0 mm	127.0 mm	-	-	-	-	-	-	-	-
5	125	133.0 mm China	-	-	-	-	-	133.0 mm**	-	-
		139.7 mm BS/ISO	DN 125	139.7 mm	139.7 mm	139.8 mm	-	139.7 mm	139.7 mm	-
-	-	141.3 mm	-	-	-	-	5"	-	-	-
-	152.4 mm	152.4 mm	-	-	-	-	-	-	-	-
6	150	159.0 mm China	-	-	-	-	-	159.0 mm	-	-
		165.1 mm JIS/BS	-	165.1 mm	-	165.2 mm	-	-	165.1 mm	-
-	-	168.3 mm	DN 150	-	DN 150	-	6"	DN 150	-	DN 150
-	6	193.7 mm	-	-	-	-	-	-	-	193.7 mm
-	203.2 mm	203.2 mm	-	-	-	-	-	-	-	-
8	200	216.3 mm JIS	-	-	-	216.3 mm	-	-	-	-
		219.1 mm	DN 200	DN 200	DN 200	-	8"	DN 200	DN 200	DN 200
-	254.0 mm	254.0 mm	-	-	-	-	-	-	-	-
10	250	267.4 mm JIS	-	-	-	267.4 mm	-	-	-	-
		273.0 mm	DN 250	DN 250	DN 250	-	10"	DN 250	DN 250	DN 250
-	304.8 mm	304.8 mm	-	-	-	-	-	-	-	-
12	300	318.5 mm JIS	-	-	-	318.5 mm	-	-	-	-
		323.9 mm	DN 300	DN 300	DN 300	-	12"	-	-	-
14	350	355.6 mm	DN 350	DN 350	DN 350	DN 350	14"	DN 350	-	-
		377.0 mm China	-	-	-	-	-	377.0 mm	-	-
16	400	406.4 mm	DN 400	DN 400	DN 400	DN 400	16"	DN 400	-	-
		426.0 mm China	-	-	-	-	-	426.0 mm	-	-
18	450	457.2 mm	DN 450	DN 450	DN 450	DN 450	18"	DN 450	-	-
		480.0 mm China	-	-	-	-	-	480.0 mm	-	-
20	500	508.0 mm	DN 500	DN 500	DN 500	DN 500	20"	DN 500	-	-
		530.0 mm China	-	-	-	-	-	530.0 mm	-	-
22	550	558.8 mm	-	-	-	DN 550	22"	559.0 mm	-	-
		580.0 mm China	-	-	-	-	-	580.0 mm	-	-
24	600	610.0 mm	DN 600	DN 600	DN 600	DN 600	24"	DN 600	-	-
		630.0 mm China	-	-	-	-	-	630.0 mm	-	-

important note:

Nominal designations are used where the actual O.D. of the pipe matches the ANSI size. Otherwise both the nominal and actual O.D. are listed.

China sizes are listed as actual O.D. in mm.

** China sizes are tubing sizes.

specifications, standards, codes & organizations

Shurjoint production facilities are certified to ISO 9001. Products are designed to conform and meet or exceed all applicable domestic and international standards and are listed, approved and or certified by various approval bodies and registration authorities. Shurjoint is also active in industry and environmental organizations.



ANSI
American National Standards Institute



ANSI/AWWA
American Water Works Association
C606 (latest edition)

ASME
American Society of Mechanical Engineers
Power Piping, B31.1
Building Services Piping, B31.9



ASTM
American Society of Testing and Materials
F 1476-01 Couplings
F 1548-01 Fittings
F 1155 Shipbuilding



CNBOP-PIB
Scientific and Research Centre for Fire Protection - National Research Institute



CSA
Canadian Standards Association B-242



DLEG
State of Michigan Board of Mechanical Rules



FM
Factory Mutual Research Corp. -
Approved for Fire Protection Services



FESC
Japan Fire Equipment Safety Center



IAPMO R&T
IAPMO Research and Testing, Inc.



LLOYD
Lloyd's Register Quality Assurance
ISO 9001:2015



LPCB
Loss Prevention Certification Board
LPS-1219



NFPA
National Fire Protection Association
NFPA 13



NSF
NSF/ANSI 61 Drinking Water System Components - Health Effects
NSF/ANSI 372 Drinking Water System Components - Lead Content



NYC MEA
New York City Department of Buildings, Material & Equipment Acceptance



NYPA
New York Power Authority



PED
Pressure Equipment Directory
2014/68/EU



UL
Underwriter's Laboratories, Inc. - UL213



ULC
Underwriter's Laboratories of Canada



TSUS
Technický a Skúšobný Ústav Stavebný, n. o.



VdS
VdS Schadenverhütung